ROLES OF CHEMISTS 12

You may need a textbook and/or dictionary to help you with this worksheet.

There are many roles that chemists play in modern society. Four main areas are:

- industry
- pure research
- education
- community work

1 Read the following passages and define each term in *italics*.

Chemists Three branches of chemistry are organic, inorganic and physical. Organic chemistry involves the study of carbon compounds; inorganic chemistry is the study of compounds not based on carbon; and physical chemistry is concerned with the physical characteristics of *matter*.

Industry In our world we depend on manufactured goods and there are many different industries that produce these goods. For example, chemists can be found working in the metal, *petrochemical*, *pharmaceutical*, agriculture, fibre, cosmetics, food and electronic industries. Industrial chemists tend to be employed in *quality control* and *research and development*. Chemists working in quality control ensure that the standard of goods produced is of the required quality. This means checking *raw materials* as they come into a factory and carrying out an *analysis* to make certain these materials are what they are supposed to be.

Chemists need to check at many stages of the manufacturing process that conditions such as temperature, concentration of solutions and pressure are within specified *margins*. Chemists are also responsible for ensuring that factory *emissions* do not exceed specified limits. Research and development (R & D) is a process that improves existing *products* and manufacturing processes, and designs new ones. The properties of materials are investigated and these materials can be altered or used in new ways. The use of computers has speeded up much of the research done by chemists and many processes have been *automated*.

Pure research Sometimes chemists investigate substances and reactions without an obvious end product for *commercial* use. This pure research can lead to discoveries that are used by other scientists to produce goods or processes that help society. Pure research is often carried out by government bodies, some large firms and *academic institutions* because it is expensive, takes many years and is generally without immediate financial benefit.

Education *Chemistry* (either as a separate subject or as part of science) is taught in almost all high schools. Students who wish to study life sciences at tertiary level are often advised to take chemistry in the senior years of high school. Universities offer many graduate and postgraduate courses in chemistry and subjects like *biochemistry* where biology and chemistry overlap.

Community work Most chemists involved in community-related work are engaged in analytical and R & D roles. The major employers include:

- government research and scientific organisations, such as CSIRO
- local government
- hospitals and health care organisations
- law enforcement agencies.

Organisations like CSIRO often take on research tasks that may not seem immediately profitable, but have the long-term goal of contributing to public welfare. Local government may be concerned with environmental issues and hospitals with public health issues. *Forensic* chemists play a role in analysing evidence in criminal cases.

2 Quality control must be assessed throughout the whole production process and not just at the raw materials stage. Suggest why.

3 Why is pure research not carried out by many organisations?

4 What is the benefit of pure research if it is not known how the findings will be applied?

Research **5** Select one industry. Describe a role played by a chemist in that industry.

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